PuKK Manual

Integration Guide

Version 1.1.0

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1. Introduction

This document describes the workings and operations for the ProDVX PuKK Workspace. This product is still in development, therefore some bugs might still be present. Please send your feedback to feedback@prodvx.com.

2. Installation

The box contains the following:

- PuKK Workspace.
- 1.5m USB Type-C cable.
- Manual Booklet.
- Mounting Screws and stickers.

Plug the USB Type C cable into the Type-C Connector on the PuKK. Ensure that a proper source is used, for example; a ProDVX USB-C Power Adapter.

The PuKK will go into its boot sequence and the LED ring will show one green quarter that indicates the boot was completed.

2.1 Mounting

The PuKK Workspace can be mounted to flat surfaces using the two supplied screws and the mounting holes in the cradle of the PuKK. Separate stickers are also supplied which can be used for mounting without creating holes.

3. Configuration

The PuKK requires configuration via NFC. To apply a configuration, a NFC card with the configuration JSON file must be held on the PUKK.

An example configuration can be found here: example_config_v1.1.0

To obtain an NFC configuration card you require the following:

- The configurator. This can be:
    - An APPC device with integrated NFC and the Pukk Companion Application. *This is the ProDVX supported solution. Note that this also requires a separate power source for the APPC, such as a USB-C Power Adapter or PoE*

connection. The PuKK can be supplied with power from the APPC with the USB cable.

- A device with an active network connection and a method of writing this configuration to an NFC card.

- An NFC Writable card (NTAG213/215/216)

3.1 Configuration Fields

A full overview of the configuration fields can be found here as well as configuration specific documentation.

3.2 With the Companion Application

The Companion Application allows you to create a configuration or import an existing configuration and write it to an NFC card. Below are the instructions for configuring your PuKK Workspace devices.

1. Open the companion application on your ProDVX Android device.
2. The application will ask for a configuration.
   A. Existing configuration
      - If you have a configuration on your device, you can upload it using the "upload" button.
      - After you have selected the configuration, the app will show it to you for review.
      - Select "Accept".
   B. Form
      - If you do not have a configuration, you can press "create a configuration".
      - The app will display a form that you can fill in for all the required information. The app will automatically check for verifiable errors.
      - When you are done, you can click the "Finish" button.
      - The app will show your configuration for a final check.
      - Select "Accept".
3. After you finish the configuration setup, tap "Write to NFC".
4. The app will show a prompt to hold your NFC card to the reader. This will write the configuration onto the card. A confirmation prompt will show if the writing process is successful.
5. When the PuKK has booted, tap the top of the PuKK with the NFC card. Ensure to leave the NFC card long enough on the PuKK to full read the configuration.
6. The PuKK led ring will start showing a rotating blue led to indicate that it is reading the configuration.

6.1 If the configuration is invalid or unreadable (due to configuration issues), the device will flash red fast and then breathe red slowly until a new read is done.

6.2 If the configuration is successful, the device will flash green fast and reboot.

7 After a reboot, the device will show which parts of the configuration were successful with the LEDs. (Explanation follows in Chapter 5. Boot Indicator)

3.3 Without the Companion Application

1. Go to https://prodvx.github.io/PuKK_Configurator/.
2. Fill in your details and download the configuration file using the “Download” button.
3. Use your preferred method to write the configuration file to your NFC card. For the best results, write is as Plain Text without spaces and new-lines.
4. When the PuKK has booted, tap the top of the PuKK with the NFC card. Ensure to leave the NFC card long enough on the PUKK to full read the configuration.
5. The PuKK LED ring will start showing a rotating blue led to indicate that it is reading the configuration.
    1. If the configuration is invalid, or unreadable (due to configuration issues), the device will flash red fast 3 times and go back with the original red flashing.
    2. If the configuration is successful, the device will flash green fast and reboot.
6. After a reboot, the device will show which parts of the configuration were successful with the LEDs (Explanation follows in Chapter 5. Boot Indicator)

3.4 Reconfiguration

For security purposes, tapping the PuKK Workspace with an NFC card with a configuration will not by default overwrite the existing configuration.

If you need to reconfigure the PuKK, power down the device, and power it back up. You need to hold the NFC card with the configuration to the PuKK within 10 seconds. This will apply the new configuration and reboot the device.

You can also remove any known information by holding the reset pin on the bottom for 5 seconds. This will perform a factory reset and restart the device.

If no configuration is already present, it will always accept a configuration.

4. Boot Indicator

Once the boot of the PuKK Workspace has been completed, the following LED quadrants indicate which parts of the configuration were successful.

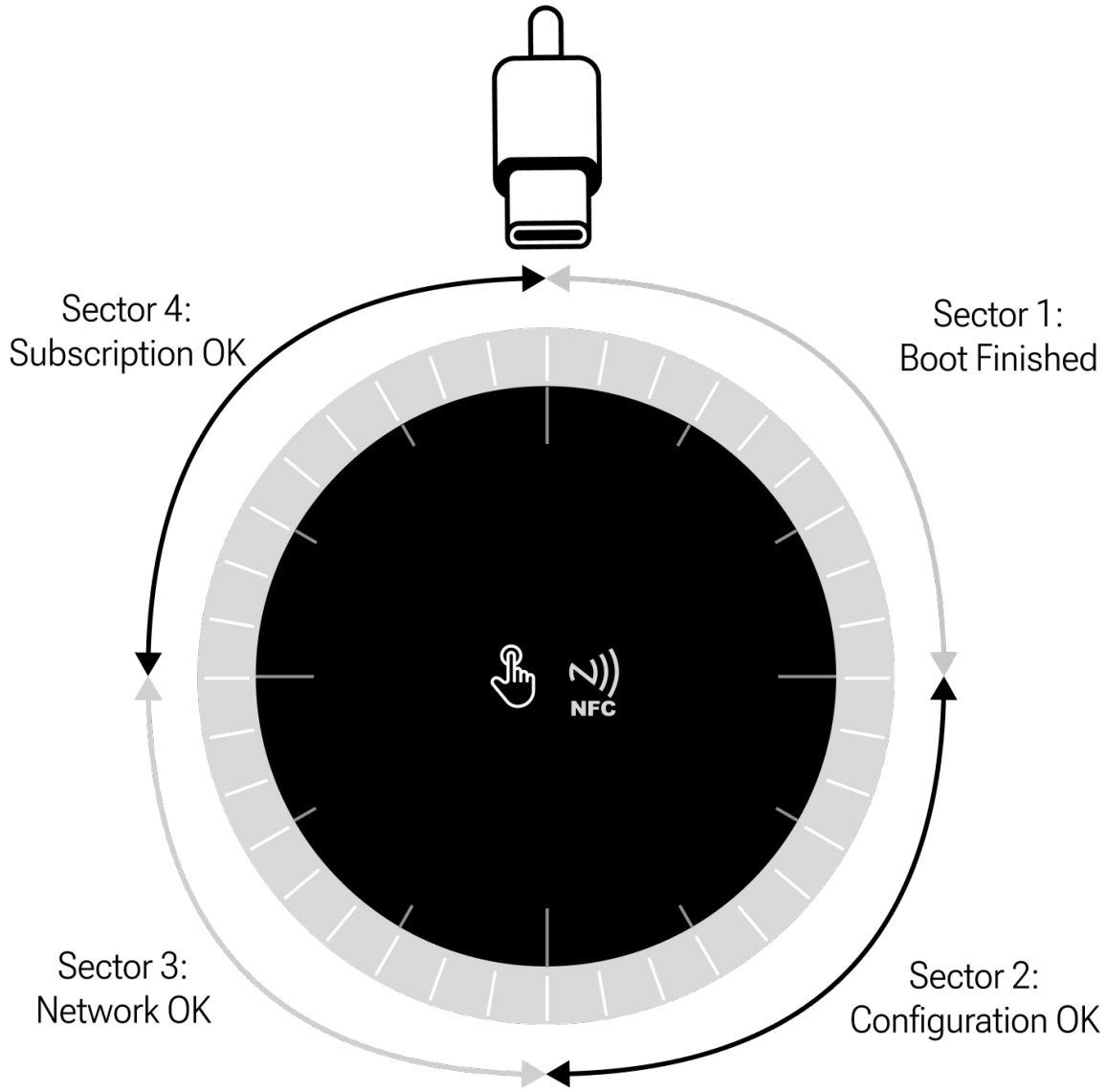


Figure 1: Boot Sectors

The sectors indicate the following information:

1) If Sector 1 is green, it indicates a finishing boot. *(Note. Currently it might take a few extra seconds to be fully booted after LEDS turn green)*

2) If Sector 2 is green, it indicates a valid configuration.

3) If Sector 3 is green, it indicates a valid network connection (received an IP-address).

4) If Sector 4 is green, it indicates that the device has a valid subscription (currently unused).

5. Subscription

To support flexible deployment and business models, the PuKK Workspace includes a built-in mechanism that verifies the validity of an active subscription through a secure server connection.

This functionality enables integration of the PuKK Workspace into a subscription-based offering, to be able to include the device in a monthly service model. By doing so, organizations can benefit from the solution without requiring an upfront investment.

The PuKK Workspace is configured with a free subscription by default. If required, it can be adjusted to support a rental or lease model. Naturally, this mechanism does not apply to one-time purchases.

This feature is currently unused in the present firmware, therefore the PuKK will always be active if configuration is completed.

6. Usage

The PuKK can be controlled utilizing HTTP(S) requests and responses and MQTT messages.

These will be explained in the following chapter.

6.1 REST API (HTTP/S)

You can directly command a PuKK using HTTP Requests by making use of the REST API. This API runs locally on the devices and can be addressed from the local network using it's IP-address.

Use the REST API Specification for detailed information about the REST API.

PuKK_REST_API_Specification

6.1.1 Authorization

Starting from 1.1.0, the PuKK will verify requests to its API. This is done using a Bearer Authorization token.

You can use the token generator on the ProMGR Portal to generate a token using "ProdvxPukk" as device key. You can set a new device key by using the "/setDeviceKey" endpoint.

6.2 HTTP/S Responses

You can also control the PuKK using HTTP responses.

The PuKK sends a request by polling (by default every 5 seconds) and on an event (button press or NFC) to the specified server address. This request can be responded to with HTTP Code 200 (OK) and a command and LED values in the response body. This will tell the PuKK how to act.

You can also respond with HTTP code 204 (No content) to tell the PuKK that no new action is required. This will tell the PuKK that it doesn't have to do anything.

Polling is set at an interval of 5 seconds by default, but can be configured.

On the very first poll request after boot, the PuKK will add the `&force=1` query parameter. This can be used after a crash or unrecoverable error to ensure that the PuKK will receive the latest information available.

For more information on Events and Polling see the following documentation.

PuKK_Event_Specification

6.3 Commands (MQTT/S)

You can also setup the PuKK over MQTT.

Using a broker, you can send an address to a specific address with a topic suffix. Check the Configuration for which parameters are available.

The PuKK will send a request upon button presses and NFC interactions to the specified broker and topic.

7. LED Operations

The LEDS can be addressed by the REST API or by responding to the requests. Find the specific requests in the REST API section 5.2 and responses in 5.3.

All colors are accessible by LRGB as “[0..100], [0..255], [0..255], [0..255]”.

The LEDS operate with a specific type. This list specifies the types that are by default integrated in the firmware.

For any LED or group of LEDs that can be directly assigned, the order is starting clockwise from North/Up position (see Figure 2)

Table 1: Led Types

Type	Operation	Values and Description
0	Off	N/A
1	On (All LEDs, static)	LRGB: Color value. Duration_ms: Time until off.
2	Marquee (Single LED rotation)	LRGB: Color value. Duration_ms: Time until off. Speed_ms: Speed of the rotation. Counter_clockwise: Flag if rotation should be counter clockwise (Default: False)
3	Breathe (Increasing and decreasing brightness over time)	LRGB: Color value of the full ring, (L is max brightness) Duration_ms: Time until off. Speed_ms: Speed of increase and decreasing brightness
4	Rainbow (Rotating rainbow gradient)	Duration_ms: Time until off. Speed_ms: speed of rotation, Counter_clockwise: Flag if rotation should be counter clockwise (Default: False)
5	Quadrant (4 individually lighted quarters)	Quadrants: LRGB Color values for each quadrant. Duration_ms: Time until off.
6	Individual (Individual LED control)	Leds: LRGB Color values for each LED. Duration_ms: Time until off.
7	Clock	Initial_Color: LRGB Initial value for all leds before clock starts. Intermediate_Color: LRGB value for the next led to go to end_color. End_color: LRGB value for all leds after clock ends. RotationTime_s: Time in seconds for one full rotation. Counter_clockwise: Flag if rotation should be counter clockwise (Default: False)

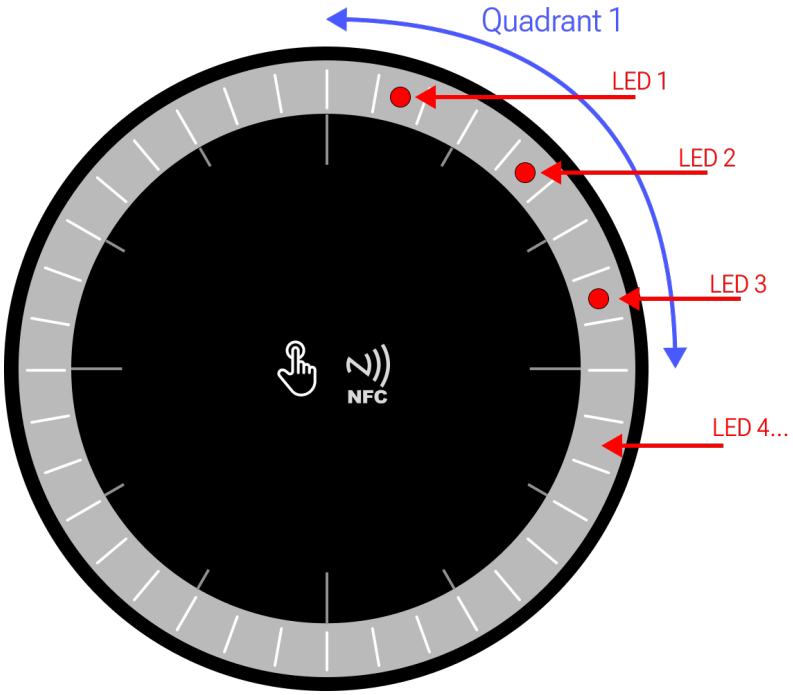


Figure 2: Led Arrangement

8. OTA

The PuKK Workspace can automatically fetch new updates when they are available. The update process can also be started manually.

8.1 OTA cycle

The PuKK makes a request to the ProDVX OTA server. This will trigger a download that contains version information about the most recent firmware. If a newer version is found, it will proceed with the download and install the new firmware. Your configuration will remain on the device.

8.2 OTA update start

If OTA updates are enabled in the configuration, the request to the OTA server is done upon boot.

If OTA is disabled, you can force the OTA request by making a REST API request to the PuKK with the /ota endpoint or by sending an ota command (MQTT or Poll response). This will start the OTA cycle. For specific uses, the location of OTA can be overridden, this is noted in the API specification.

8.3 Manual update

At the current moment, manual burn of the PuKK Workspace firmware is unsupported.

8.4 OTA Server information

The OTA service is hosted by Hetzner. Hetzner is a German-based web hosting provider and data center operator known for offering highly reliable and cost-effective infrastructure. It provides dedicated servers, cloud hosting (VPS), managed servers, and colocation services from data centers in Europe (Germany, Finland) and the USA.

The specific files are hosted in Falkenstein, Germany.

9. Other useful information

9.1 mDNS

The PuKK is also broadcasting an mDNS hostname. Use ping <hostname> to find the IP address when connected to the same network.

FW Version	hostname	Description
1.1.0 or lower	esp-home.local	-
1.1.1 or higher	pukk-XXXXXX.local	XXXXX: the last 6 characters of the MAC address

10. FAQ

Q: What happens when the HTTP CMS server is unavailable?

When the HTTP CMS server is unreachable, the PuKK Workspace will retain its last given instructions. It will keep retrying the endpoint.

Since the default behavior on button press makes the LEDs a light yellow, this will also stay in place.

11. Troubleshooting

Please contact support@prodvx.com for detailed support about the PuKK Workspace.

12. Version History

Version	Author	Date
1.0	ProDVX Software	01/03/2026
1.1.0	ProDVX Software	27/10/2026